Comprehensive Audited Analysis of seq2seq.ipynb

This report audits every code cell in the notebook **seq2seq.ipynb**, maps results to their exact cell locations, compares "With Attention" vs. "Without Attention" runs across both $t+1$ and horizon-wide scopes, extracts actionable development insights, and provides recommendations for next steps.

Note: All tables below are ranked in increasing order of features used, with Attention (Yes) listed before Standard (No) for each feature size.

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1. Cell-by-Cell Notebook Audit

Here is the exact description and mapping of all 12 code cells in `seq2seq.ipynb`:

Cell Index	Label / Purpose	Model Type	Feature Set Description	Training Loss (Epoch 20)	Val Loss (Epoch 20)	Output Status
Cell 7	Refined Attention	Univariate Attention (Instance 2)	`Total Load (MW)` (1 feature)	0.1100	0.0915	Success (Mean MAE: 7.94 MW)
Cell 2	Baseline Attention	Univariate Attention (Instance 1)	`Total Load (MW)` (1 feature)	0.1157	0.0995	Success (Mean MAE: 9.08 MW)
Cell 8	5-Feature Attention	Multivariate Attention	Load + Cyclical Time (`hour`/`dow`)	0.0670	0.1144	Success (Mean MAE: 10.01 MW)
Cell 3	5-Feature Standard	Multivariate Standard	Load + Cyclical Time (`hour`/`dow`)	0.0721	0.1540	Success (Mean MAE: 10.26 MW)
Cell 9	7-Feature Attention	Multivariate Attention	Load + Cyclical + Weather (`temp`/`rh`)	0.0499	0.2117	Success (Mean MAE: 10.85 MW)
Cell 4	7-Feature Standard	Multivariate Standard	Load + Cyclical + Weather (`temp`/`rh`)	0.0554	0.2100	Success (Mean MAE: 11.35 MW)
Cell 10	8-Feature Attention	Multivariate Attention	Load + Lags (24h/48h/168h) + Weather + Month	0.0569	0.1419	Success (Mean MAE: 10.04 MW)
Cell 5	8-Feature Standard	Multivariate Standard	Load + Lags (24h/48h/168h) + Weather + Month	0.0610	0.2877	Success (Mean MAE: 12.03 MW)
Cell 11	14-Feature Attention	Multivariate Attention	Full Feature Set (14 features)	0.0418	0.1735	Success (Mean MAE: 11.52 MW)
Cell 6	14-Feature Standard	Multivariate Standard	Full Feature Set (14 features)	0.0486	0.2854	Success (Mean MAE: 11.58 MW)
Cell 0	Drive Mount	*None*	Colab environment initialization	*N/A*	*N/A*	Success (Mounted)
Cell 1	Feature Analysis	*None*	Mutual Information & Correlation	*N/A*	*N/A*	FileNotFoundError (Colab drive path mismatch)

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2. Model Performance Comparisons

Table A: First-Hour ($t+1$) Prediction Metrics

This table evaluates how well the models forecast the immediate next hour, mapping train/val losses and assessing overfitting levels.

Model Configuration	Attention ?	Final Train Loss (MSE)	Final Val Loss (MSE)	Test MAE ($t+1$)	Test RMSE ($t+1$)	Test R^2 ($t+1$)	Overfitting Verdict
Univariate (1 Feature - Cell 7)	**Yes**	0.1100	0.0915	**6.56 MW**	**9.08 MW**	**0.9244**	**Highly Stable.** Best convergence.
Univariate (1 Feature - Cell 2)	**Yes**	0.1157	0.0995	7.01 MW	9.56 MW	0.9163	**Stable.** Slightly higher initial loss.
Univariate (1 Feature - Local)	No	0.1062	0.1025	6.61 MW	8.82 MW	0.9286	**Highly Stable.** Closely matched train/val.
Multivariate (5 Features - Cell 8)	**Yes**	0.0670	0.1144	8.17 MW	11.46 MW	0.8797	**Stable.** Overfitting is minimized.
Multivariate (5 Features - Cell 3)	No	0.0721	0.1540	8.78 MW	12.84 MW	0.8488	**Moderate Overfitting.** Val loss climbs.
Multivariate (7 Features - Cell 9)	**Yes**	0.0499	0.2117	9.16 MW	13.00 MW	0.8450	**Severe Overfitting.** Weather features add noise.
Multivariate (7 Features - Cell 4)	No	0.0554	0.2100	9.29 MW	12.32 MW	0.8610	**Severe Overfitting.** Weather features add noise.
Multivariate (8 Features - Cell 10)	**Yes**	0.0569	0.1419	8.73 MW	12.31 MW	0.8612	**Moderate Overfitting.** Attention stabilizes lags.
Multivariate (8 Features - Cell 5)	No	0.0610	0.2877	10.01 MW	14.23 MW	0.8145	**Extreme Overfitting.** Diverges heavily.
Multivariate (14 Features - Cell 11)	**Yes**	0.0418	0.1735	9.38 MW	13.09 MW	0.8430	**Severe Overfitting.** High input dimensionality.
Multivariate (14 Features - Cell 6)	No	0.0486	0.2854	9.41 MW	13.19 MW	0.8405	**Extreme Overfitting.** Diverges heavily.

Table B: Pairwise Full-Horizon ($t+1$ to $t+24$) Average Metrics

This table evaluates average performance across the entire 24-hour day-ahead horizon, showcasing the direct impact of adding the Attention layer.

Input Feature Configuration	Attention?	Final Train Loss (MSE)	Final Val Loss (MSE)	Mean MAE (MW)	Mean RMSE (MW)	Overall R^2 Score	Overfitting Ratio (Val/Train Loss)	Actionable Insights & Direct Verdict
Univariate (1 Feature - Cell 7)	**Yes**	0.1100	0.0915	**7.94**	11.77	0.8712	**0.83 (Stable)**	**Best Model.** High stability, zero overfitting. Use this univariate model as the current production baseline.
Univariate (1 Feature - Local)	No	0.1062	0.1025	8.10	11.68	0.8731	0.96 (Stable)	**Stable Baseline.** Use if attention context overhead is an issue.
Multivariate (5 Features - Cell 8)	**Yes**	0.0670	0.1144	10.01	14.91	0.7938	**1.70 (Stable)**	**Attention mitigates cyclical noise.** Cyclical features in encoder add noise without future context.
Multivariate (5 Features - Cell 3)	No	0.0721	0.1540	10.26	14.98	0.7928	2.13 (Moderate)	**Moderate Overfitting.** Diverges over the horizon.
Multivariate (7 Features - Cell 9)	**Yes**	0.0499	0.2117	10.85	15.75	0.7707	4.24 (Severe)	**Past Weather is Noise.** Adding past weather to encoder degrades accuracy. Switch weather to decoder forecasts.
Multivariate (7 Features - Cell 4)	No	0.0554	0.2100	11.35	15.82	0.7685	3.79 (Severe)	**Severe Overfitting.** Encoder struggles with high-dim noise.
Multivariate (8 Features - Cell 10)	**Yes**	0.0569	0.1419	10.04	14.99	0.7921	**2.49 (Moderate)**	**Attention rescues lags.** Attention cuts error and val loss in half when lag dimensions are high.
Multivariate (8 Features - Cell 5)	No	0.0610	0.2877	12.03	17.88	0.7043	4.71 (Extreme)	**Bottleneck collapse.** Strong lag signals cannot pass through standard encoder context.

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Multivariate (14 Features - Cell 11)	**Yes**	0.0418	0.1735	11.52	17.15	0.7278	4.15 (Severe)	**Extreme feature redundancy.** Even with attention, high feature counts lead to severe overfitting.
Multivariate (14 Features - Cell 6)	No	0.0486	0.2854	11.58	16.69	0.7419	5.87 (Extreme)	**Severe overfitting.** High-dimensional parameters diverge.

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3. Key Actionable Insights

Insight 1: Stop Feeding Historical Weather to the Encoder

- **Observation**: The 7-Feature model (Load + Past Weather) degrades overall horizon R^2 to **0.7707** compared to **0.8712** for Univariate.
- **Why**: The grid load at $t+12$ depends on the temperature at $t+12$, not the temperature 3 days ago. Forcing the LSTM encoder to process historical weather adds dimensionality noise.
- **Action Item**: Remove weather variables from the encoder sequence. Future weather forecasts must be fed directly to the decoder.

Insight 2: Enable Attention whenever Lags are Present

- **Observation**: Without attention, adding lag features ($t-24, t-48, t-168$) degrades the model's MAE to **12.03 MW**. With attention, MAE recovers to **10.04 MW**.
- **Why**: Lags create strong temporal pathways that get lost in the standard encoder's single context vector. Attention allows the decoder to align directly with these lagged intervals.
- **Action Item**: Never use multivariate lags in standard Seq2Seq models without an Attention layer.

Insight 3: Transition to a "Future-Guided" Decoder

- **Observation**: Univariate Attention remains the best model overall because the multivariate models lack tomorrow's schedule context during decoder generation.
- **Action Item**: Restructure the inputs so that tomorrow's hourly temperature forecasts and calendar markers are fed directly into the decoder step-by-step during prediction.